

GVARV, Norway

WMO: 014700

Lat: 59.383N

Lon: 9.201E

Elev: 93

StdP: 100.21

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.5	-13.2	-19.7	0.7	-3.8	-16.8	0.9	-8.9	7.0	1.7	6.2	2.0	2.0	50	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	26.6	17.4	24.6	16.0	22.7	15.0	18.4	24.5	17.4	22.4	16.5	20.8	2.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	11.6	19.9	15.4	11.0	19.1	14.6	10.5	18.3	52.1	24.5	49.1	22.5	46.5	20.7	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.1	5.2	4.4	DB	-16.8	29.7	2.6	1.6	-18.7	30.9	-20.2	31.8	-21.7	32.7	-23.5	33.9	(4)
(5)			WB	-17.0	19.9	2.8	0.8	-19.0	20.5	-20.7	20.9	-22.2	21.4	-24.3	21.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-3.0	-2.3	1.9	6.6	10.6	14.8	17.2	15.8	11.9	6.4	1.9	-2.1	(6)	
	DBStd	7.97	4.60	4.66	4.10	3.27	3.54	2.60	2.78	2.25	2.72	3.37	3.61	5.33	(7)	
	HDD10.0	1899	404	344	253	112	33	1	0	0	13	119	242	377	(8)	
	HDD18.3	4287	663	577	511	353	239	111	57	84	194	371	492	635	(9)	
	CDD10.0	686	0	0	1	9	53	144	223	178	69	7	1	0	(10)	
	CDD18.3	32	0	0	0	0	2	4	22	4	0	0	0	0	(11)	
	CDH23.3	304	0	0	0	0	19	63	191	31	0	0	0	0	(12)	
	CDH26.7	46	0	0	0	0	2	9	34	1	0	0	0	0	(13)	
(14) Wind	WSAvg	1.9	2.0	2.0	2.2	2.1	2.1	2.0	1.7	1.6	1.8	1.8	2.0	2.0	(14)	
(15) Precipitation	PrecAvg	954	80	48	44	53	71	85	94	113	100	104	97	64	(15)	
	PrecMax	1436	182	150	119	130	202	171	232	185	275	255	419	116	(16)	
	PrecMin	742	12	11	2	7	7	4	21	7	38	14	14	11	(17)	
	PrecStd	154	43	32	28	34	43	42	53	47	52	56	79	29	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	9.0	15.7	20.3	25.8	28.0	29.7	26.2	21.9	16.0	12.3	10.2	(19)	
		MCWB	5.5	4.7	8.2	11.4	15.8	15.3	19.0	18.0	15.1	11.4	9.3	7.6	(20)	
	2%	DB	5.8	6.8	12.6	18.0	22.6	25.0	27.5	23.9	20.1	14.2	10.5	7.9	(21)	
		MCWB	3.6	3.3	7.4	10.3	13.4	14.4	18.5	16.3	14.2	11.0	8.9	6.0	(22)	
	5%	DB	4.6	5.3	10.0	15.4	20.0	22.9	25.6	22.2	18.4	13.0	8.9	6.5	(23)	
		MCWB	2.7	2.7	5.3	9.1	12.5	13.7	17.6	15.4	13.2	10.5	7.5	4.8	(24)	
	10%	DB	3.1	3.8	7.8	12.9	17.5	21.0	23.6	20.8	16.6	11.7	7.4	4.9	(25)	
		MCWB	1.6	2.1	3.8	7.3	11.1	13.0	16.3	14.9	12.6	9.5	6.2	3.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	5.6	9.5	12.4	16.6	17.4	20.0	18.8	16.7	12.8	10.6	8.1	(27)	
		MCDB	7.4	7.8	14.7	18.9	23.7	24.1	27.3	24.3	20.4	14.5	11.2	10.0	(28)	
	2%	WB	4.4	4.2	7.5	10.7	14.9	16.2	19.0	17.6	15.3	11.9	8.9	6.4	(29)	
		MCDB	5.5	6.0	12.3	16.0	20.1	22.3	26.0	21.7	17.9	13.3	9.7	7.6	(30)	
	5%	WB	3.0	3.1	5.8	9.5	13.2	15.2	18.1	16.7	14.4	11.0	7.7	5.0	(31)	
		MCDB	4.2	4.8	9.4	14.4	18.3	20.3	24.3	20.4	16.9	12.3	8.8	6.2	(32)	
	10%	WB	1.6	2.1	4.2	8.3	11.8	14.3	17.2	15.8	13.6	9.9	6.3	3.4	(33)	
		MCDB	2.8	3.6	7.0	12.4	16.6	19.1	22.0	19.2	16.0	11.4	7.3	4.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	6.1	8.9	9.8	9.9	10.3	9.7	9.5	8.8	6.9	4.9	4.9	(35)	
		MCDBR	6.2	8.7	12.4	14.3	13.8	13.9	13.2	11.6	11.4	8.5	7.3	6.7	(36)	
	5% WB	MCWBR	5.1	5.9	7.8	7.9	6.7	7.0	6.8	6.1	6.3	5.7	5.9	5.5	(37)	
		MCWBR	5.9	7.0	11.7	12.9	11.9	11.5	12.1	9.7	9.1	7.0	6.6	6.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.257	0.295	0.321	0.364	0.361	0.350	0.371	0.365	0.336	0.321	0.278	0.232	(40)	
		taud	2.218	2.297	2.322	2.323	2.425	2.476	2.435	2.444	2.520	2.481	2.312	2.144	(41)	
	Ebn at Noon		556	723	815	838	869	884	854	830	803	696	530	442	(42)	
		Edh at Noon	42	67	88	106	103	99	101	93	74	57	39	30	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.34	0.99	2.51	4.00	4.90	5.62	4.93	3.89	2.54	1.18	0.44	0.22	(44)	
	RadStd		0.07	0.21	0.22	0.57	0.60	0.45	0.63	0.29	0.35	0.22	0.12	0.02	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (51 neighbors)		N/A	N/A	+1.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page